Table 1. Octinoxate: MM/GBSA binding free energy, ΔG (kcal/mol), and ligand retention (%) from 20 ns MD started at each folding model's co-folded pose.

Peptide	Boltz / BoltzGen co-fold	OpenFold3 co-fold	RoseTTAFold3 co-fold
orig_f12	−13.71 (77.3%) ↑	−5.90 (7.4%) ↑	−19.51 (99.9%) ↑
s2_esm2_control	−16.25 (59.6%) •	−10.21 (33.9%) ↓	−14.98 (57.2%) ↓
s3_esm2_f4	−21.08 (99.6%) •	−13.76 (84.8%) ↑	−16.18 (88.2%) ↑
s3_orig_f12	−24.33 (100.0%) ↑	−10.10 (44.8%) ↑	−12.10 (68.9%) ↑
shuffle_control	−15.13 (75.9%) ↑	−9.39 (25.7%) ↓	−11.13 (38.7%) •
shuffle_control_esm0	−14.32 (80.2%) ↓	−15.62 (92.5%) ↓	−21.53 (99.3%) •
bg33_1	−11.31 (14.5%) ↓	−14.90 (56.4%) ↓	−12.43 (68.3%) ↓
bg33_2	−8.02 (50.9%) ↑	−11.58 (70.2%) •	−11.52 (58.7%) ↓
bg33_3	−11.86 (41.0%) ↓	−10.10 (14.9%) •	−12.44 (1.9%) •
bg33_4	−19.66 (100.0%) ↓	−17.11 (93.0%) ↓	−16.13 (81.2%) ↑

Values are ΔG in kcal/mol with retention in parentheses. ΔG is the MM/GBSA binding free energy over the whole 20 ns production run; retention is the percentage of the run with the ligand within 10 Å of the peptide. The trailing marker compares the 20 ns ΔG with the ΔG over the first 10 ns: • stable (within 0.5 kcal/mol), ↓ still decreasing (binding strengthening), ↑ still increasing (binding weakening). The Boltz / BoltzGen column is the original co-fold: BoltzGen for the bg* peptides, Boltz for the rest.

Table 2. Oxybenzone: MM/GBSA binding free energy, ΔG (kcal/mol), and ligand retention (%) from 20 ns MD started at each folding model's co-folded pose.

Peptide	Boltz / BoltzGen co-fold	OpenFold3 co-fold	RoseTTAFold3 co-fold
ox1_esm1_f8	−10.70 (75.0%) ↑	−17.95 (100.0%) ↑	−12.12 (35.2%) ↓
ox1_orig_f12	−10.31 (49.1%) ↑	−3.96 (20.5%) ↑	−12.27 (89.1%) ↓
ox2_esm1_f8	−20.81 (100.0%) ↑	−16.68 (93.2%) ↑	−10.90 (90.4%) ↑
ox2_orig_f8	−24.21 (100.0%) •	−7.23 (49.5%) ↑	−4.95 (19.6%) ↑
ox2_shuffle	−5.30 (7.1%) •	−2.98 (14.8%) ↑	−13.35 (51.0%) •
ox2_shuffle_esm0	−12.04 (94.7%) •	−0.92 (1.1%) •	−11.44 (32.6%) •
ox3_orig_f8	−18.84 (100.0%) ↑	−2.06 (2.4%) ↓	−7.41 (45.6%) ↓
bgox31_2	−10.89 (33.7%) ↑	−7.45 (65.9%) ↑	−9.06 (84.6%) ↓
bgox31_3	−15.44 (99.7%) ↑	−5.82 (15.8%) ↑	−14.36 (99.2%) •
bgox31_4	−14.05 (100.0%) ↑	−5.74 (17.9%) ↑	−11.02 (72.0%) •
bgox31_5	−16.78 (100.0%) •	−11.55 (70.7%) ↓	−14.69 (96.4%) ↑

Values are ΔG in kcal/mol with retention in parentheses. ΔG is the MM/GBSA binding free energy over the whole 20 ns production run; retention is the percentage of the run with the ligand within 10 Å of the peptide. The trailing marker compares the 20 ns ΔG with the ΔG over the first 10 ns: • stable (within 0.5 kcal/mol), ↓ still decreasing (binding strengthening), ↑ still increasing (binding weakening). The Boltz / BoltzGen column is the original co-fold: BoltzGen for the bg* peptides, Boltz for the rest.